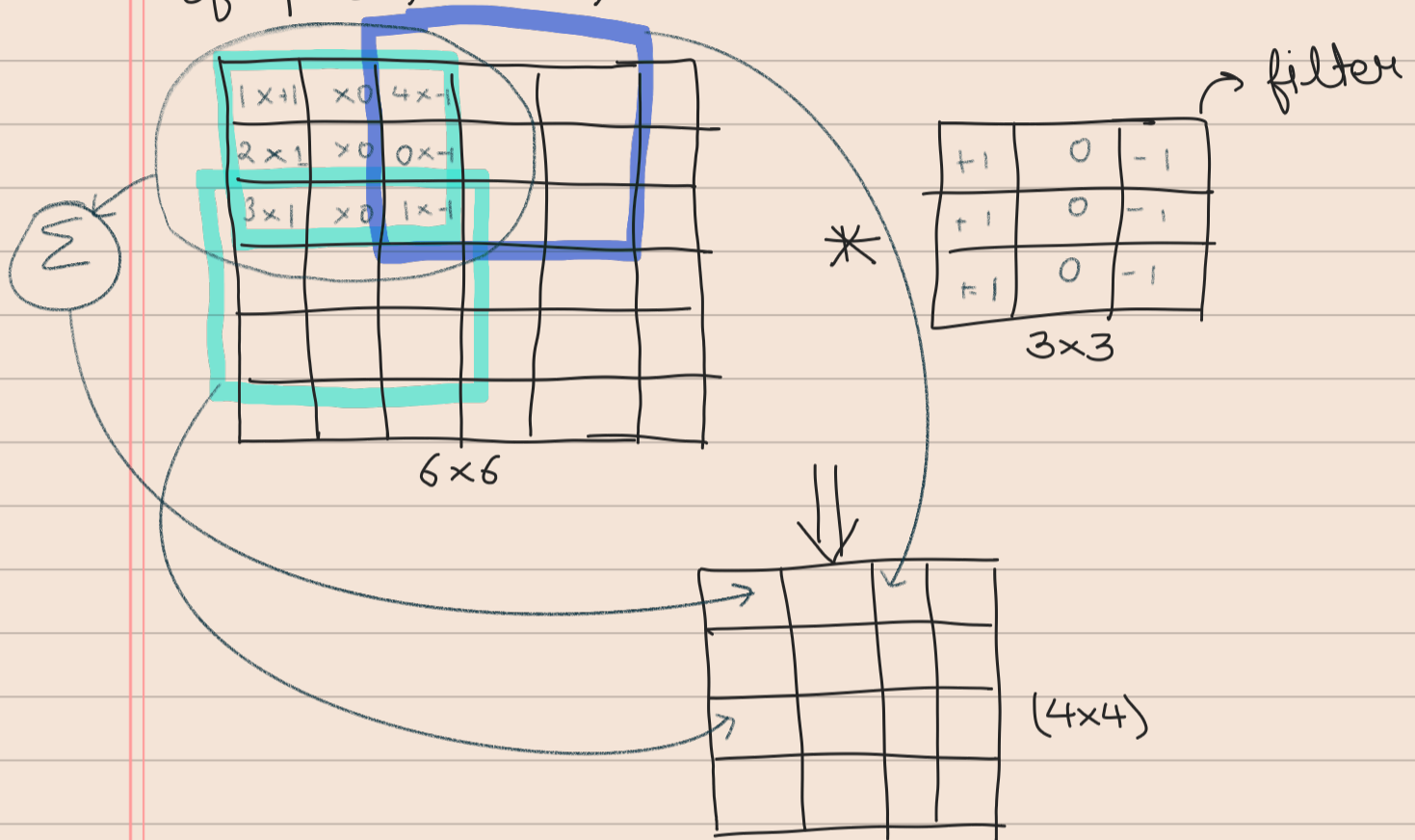


CNN:-

→ Normal NN will have so much data to handle to reduce it, we implement a sliding window across our original image grid that will reduce our number of pixels/values/data.



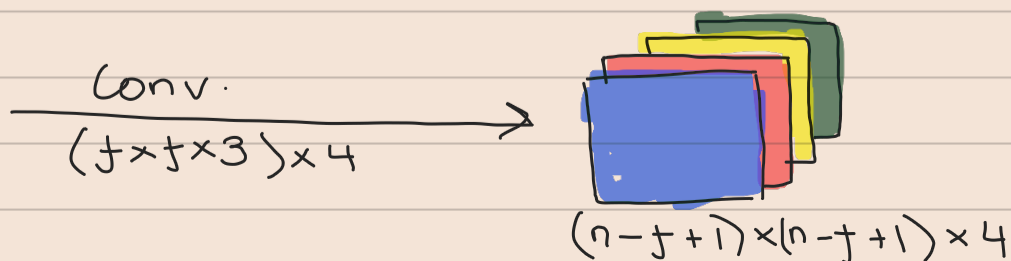
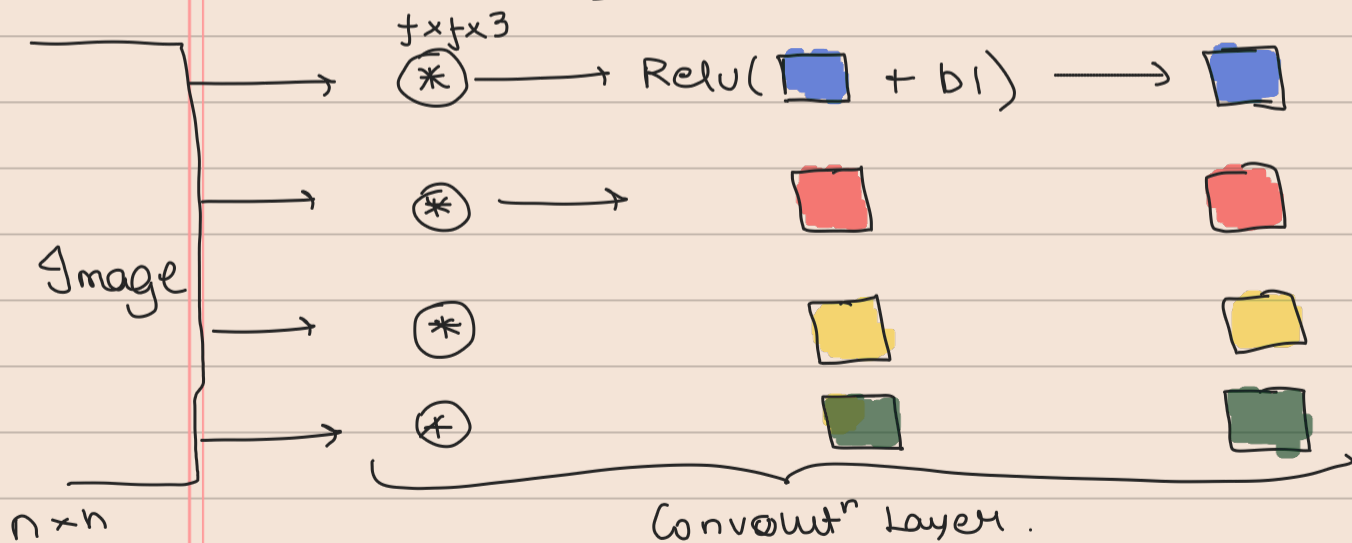
→ When any image of size $(n \times n)$ is convolved by a grid of $(f \times f)$ it reduces to $(n-f+1) \times (n-f+1)$

→ different filter when applied gives diff. info about the image, like $\begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$ gives info about vertical edge.

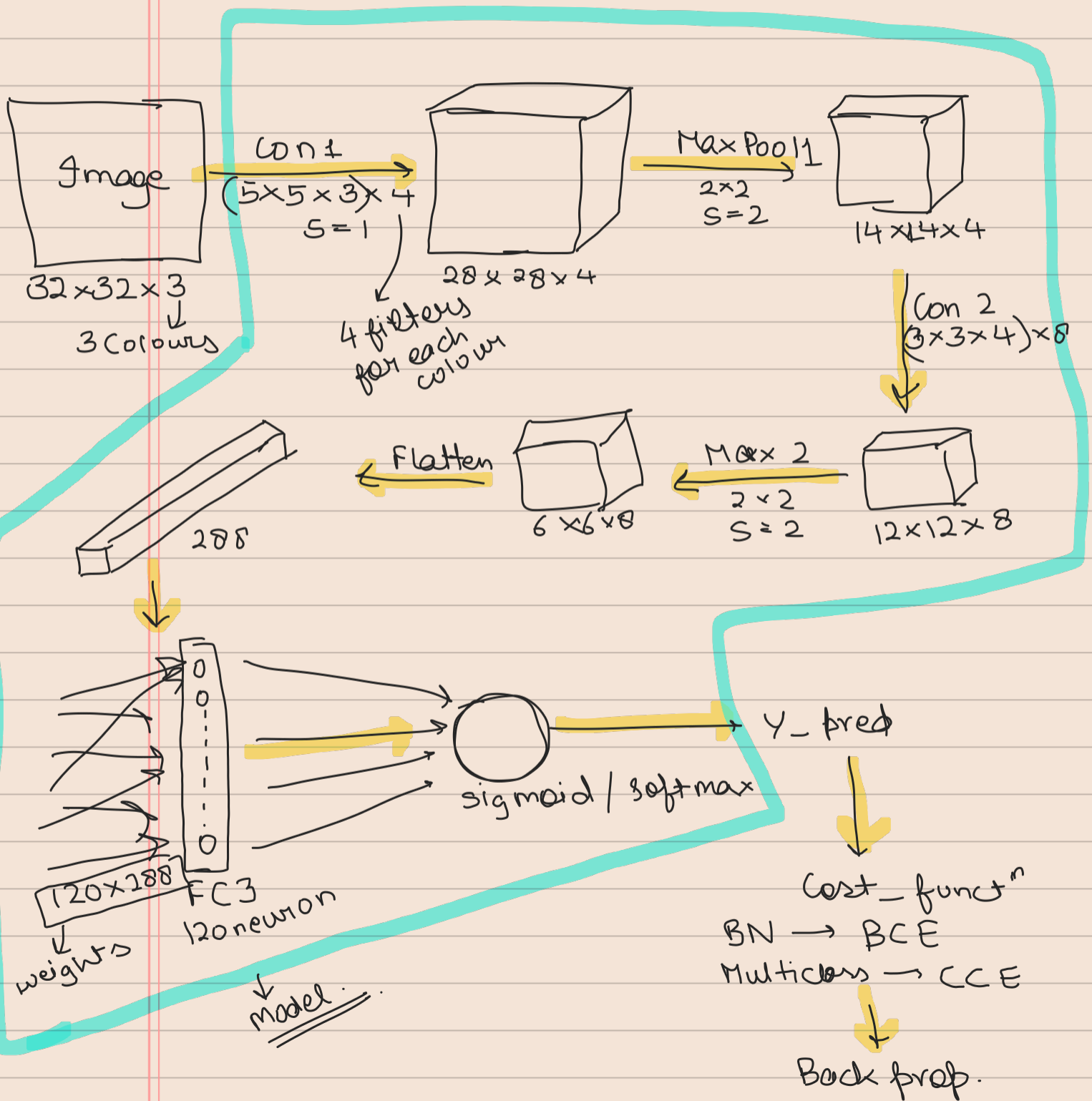
→ For coloured image we convert our image to 3D that is we split our image in diff. colour hence use 3D filter.

- padding of image to reduce biasness happened with corner & center pixel.
- Stride = 2 → moving to the 2nd positⁿ
- Max pooling
 - filter → $n \times n$
 - Stride → n
 - not necessary to use.
 - Reduces image size to Reduce computational cost, without losing data, also enhances features
- Avg Pooling → Same thing but just avg.
- Fully - Connected layers.

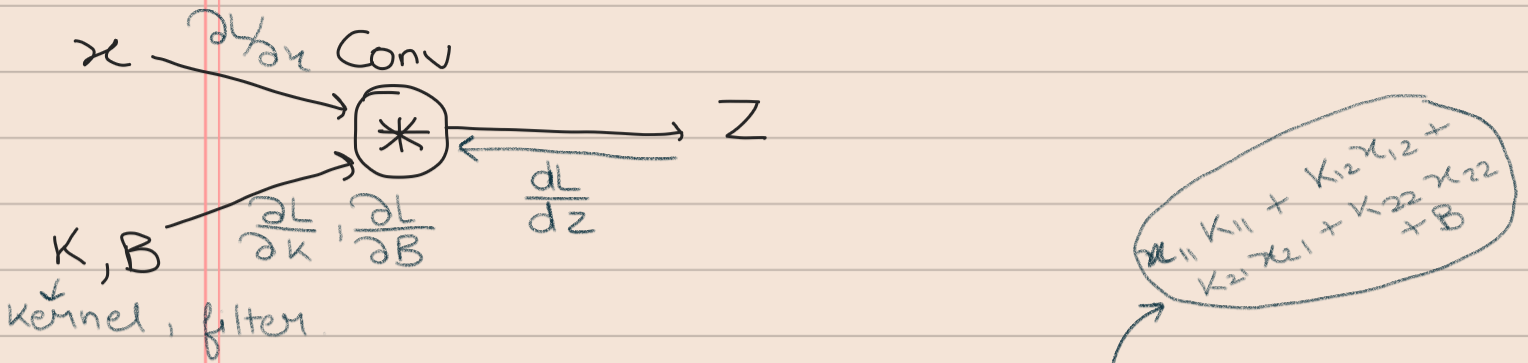
CNN Arch



CNN - Structure



Back Propagatⁿ in CNN



$$\begin{bmatrix} x_{11} & x_{12} & x_{13} \\ \vdots & \vdots & \vdots \\ \dots & \dots & x_{33} \end{bmatrix} \odot \begin{bmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix} + B = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}$$

$$\frac{\partial L}{\partial K_{mn}} = \sum \left(\frac{\partial L}{\partial Z_{ij}} \times \frac{\partial Z_{ij}}{\partial K_{mn}} \right)$$

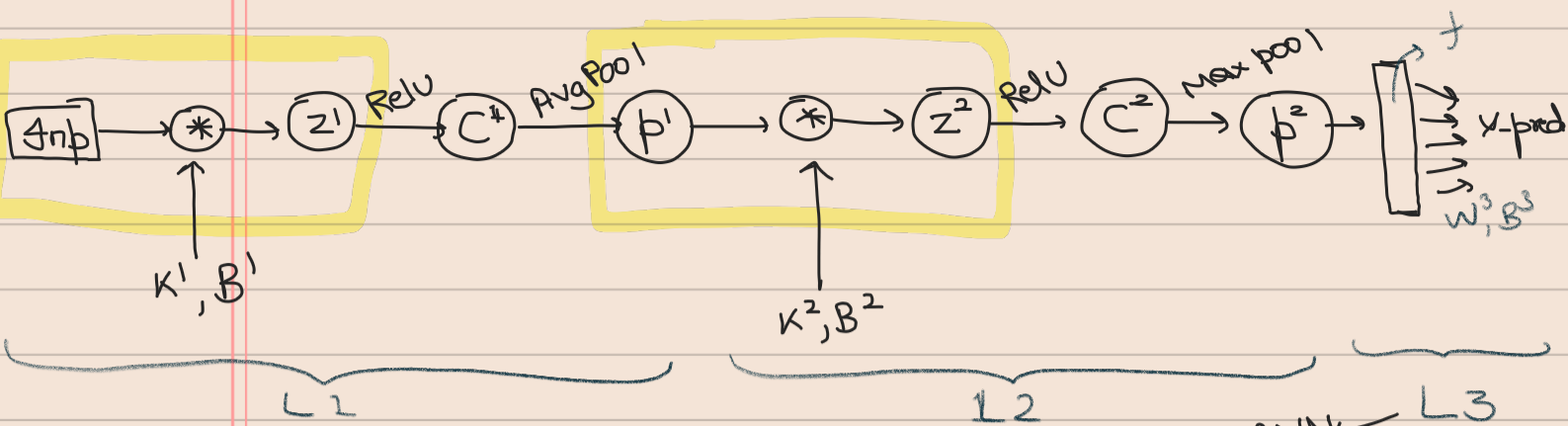
$$\begin{bmatrix} x_{11} & x_{12} & x_{13} \\ \vdots & \vdots & \vdots \\ \dots & \dots & x_{33} \end{bmatrix} \odot \begin{bmatrix} \frac{\partial L}{\partial Z_{11}} & \frac{\partial L}{\partial Z_{12}} \\ \frac{\partial L}{\partial Z_{21}} & \frac{\partial L}{\partial Z_{22}} \end{bmatrix} = \begin{bmatrix} \frac{\partial L}{\partial K_{11}} & \frac{\partial L}{\partial K_{12}} \\ \frac{\partial L}{\partial K_{21}} & \frac{\partial L}{\partial K_{22}} \end{bmatrix} = \frac{\partial L}{\partial K}$$

$$\frac{\partial L}{\partial B} = \sum \left(\frac{\partial L}{\partial Z_{ij}} \times \frac{\partial Z_{ij}}{\partial B} \right) = \sum \frac{\partial L}{\partial Z_{ij}}$$

$$\frac{\partial L}{\partial x_{mn}} = \sum \left(\frac{\partial L}{\partial Z_{ij}} \times \frac{\partial Z_{ij}}{\partial x_{mn}} \right) =$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{\partial L}{\partial Z_{11}} & \frac{\partial L}{\partial Z_{12}} & 0 \\ 0 & \frac{\partial L}{\partial Z_{21}} & \frac{\partial L}{\partial Z_{22}} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \odot \begin{bmatrix} K_{22} & K_{21} \\ K_{12} & K_{11} \end{bmatrix} = \frac{\partial L}{\partial x}$$

→ padded



$$\text{Cost} = \frac{1}{n} \sum_i^m \left(y_i \cdot \log(a_i) + (1-y_i) \cdot \log(1-a_i) \right)$$

actual

$$dP^2 = \frac{dL}{dP^2} = df \cdot \text{reshape}(P^2 \cdot \text{shape})$$

$$dC_{mn}^2 = \frac{\partial L}{\partial C_{mn}^2} = \begin{cases} \frac{\partial L}{\partial P_{xy}^2} & \text{if } C_{mn} \text{ is the max element} \\ 0 & \text{otherwise} \end{cases}$$

$$\frac{dC^2}{dZ_{mn}^2} = \begin{cases} 1 & \text{if } Z_{mn} > 0 \\ 0 & \text{if } Z_{mn} < 0 \end{cases}$$

Update $\rightarrow$

$$W_3 = W_3 - \alpha * \frac{\partial L}{\partial W_3}$$

B_3
 K_2
 B_2
 K_1
 B_1

